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Book Review Essay

The case against neuroplastic analysis: A further illustration of the irrelevance of neuroscience to psychoanalysis through a critique of Doidge's *The Brain that Changes Itself*

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In a previous paper, we critically examined claims put forth by proponents of the growing neuropsychanalytic trend (Blass and Carmeli, 2007). In particular, we called into question the claim that neuroscientific findings are relevant and important for the development and justification of psychoanalytic theory and practice. We pointed to some of the intuitions that have led to the popularity of such neuropsychanalytic claims, and then exposed the fallacies that underlie both the claims and the intuitions.

We also showed how the neuropsychanalytic trend modifies the basic aims of the analytic process, shifting away from the unique psychoanalytic concern with the understanding of meanings and the role of interpersonal discourse in discerning and justifying these meanings. As such it was having a negative impact on the development of psychoanalytic thinking and practice.

The present paper expands on and further grounds these conclusions. It responds to claims put forth in the neuropsychanalytic literature regarding an area of neuroscientific interest that we did not address in our previous work. This area of interest is the brain's plasticity (neuroplasticity). In short, brain plasticity denotes the capacity of the brain's structure and function to be altered by experience and input from the environment. Contemporary investigators of this plasticity offer evidence regarding the various forms of change that the brain remains open to throughout life, even in adulthood. It is often stressed that these findings challenge a view of the brain as immutable after a critical period during early childhood (Rakic, 2002).

Evidence associated with the notion of the brain's plasticity has recently been presented as supporting the idea that neuroscience can significantly influence psychological understanding and theoretical development. It has also been claimed that neuroplasticity is very relevant to psychoanalysis. Here, we show that the evidence associated with neuroplasticity in fact demonstrates the opposite: psychological understanding and theoretical development are in no way significantly influenced by the evidence marshalled to

support the notion of neuroplasticity, the notion has no real relevance to psychoanalysis, and in fact thinking in terms of the brain's plasticity is detrimental to the development and understanding of an analytic process.

The present study is based on an examination of the line of argument put forth in the widely acclaimed and best-selling book *The Brain that Changes Itself*. The book was written by Canadian analyst and psychiatrist Norman Doidge (2007). While not a scientific publication *per se*, it has been cited and reviewed in analytic contexts and has been influential in shaping common opinion of the relationship between psychoanalysis and neuroscience (as evidenced, for example, by the fact that in 2008 Norman Doidge received the prestigious Sigourney Award).

The brain that changes itself

Doidge's book is yet another in the flood of recently published texts on the relationship between neuroscience and mental activity (e.g. Baron-Cohen, 2011; Churchland, 2011; Damasio, 2010; Ramachandran, 2011; Shermer, 2011). The book focuses primarily on the possibility of rehabilitation of people who for a variety of reasons have been seriously damaged in terms of their daily functioning. It tells the gripping narratives of patients and professionals who have heroically and creatively struggled – and to a considerable extent succeeded – to overcome the limitations with which they were faced. These are stories of strong-willed patients who fought to re-establish normal lives despite their supposedly hopeless prognoses, and of visionary researchers who in the face of common wisdom and practice maintained that the brains of these patients could change and thus rehabilitation could be brought about.

Into these stories, Doidge weaves neuroscientific empirical findings and theoretical considerations. Thus, while reading of truly amazing accounts of cure and rehabilitation, we are also told of how these breakthrough processes of change are a consequence of the understanding of the brain's previously unfathomed plasticity. In this context Doidge writes:

The idea that the brain can change its own structure and function through thought and activity is, I believe, the most important alteration in our view of the brain since we first sketched out its basic anatomy and the workings of its basic component, the neuron. Like all revolutions, this one will have profound effects...

(pp. ix–xx)

Some of the book's chapters deal with astonishing accounts of physical rehabilitation: the rehabilitation of balance in a woman with inner ear damage (Chapter 1), the rehabilitation of functions that were lost due to stroke (Chapter 5) and the overcoming of phantom pain among amputees (Chapter 7). Other chapters are focused on cognitive rehabilitation: rehabilitation in cases in which intellectual, linguistic and learning capacities were damaged (Chapters 2, 3, 11) or in which elderly individuals could preserve and even improve their cognitive abilities despite the aging process they were undergoing (Chapter 10). Other chapters deal with emotional and impulsive disorders and how these and their debilitating effects are dramatically overcome (Chapters 4, 6).

In all these instances, alongside the stories, Doidge introduces underlying neuroscientific explanations. The reader is invited to consider the value of the dialogue to be had between the brain sciences and a range of therapeutic fields, physical and psychical in nature. Doidge also acquaints us with a wide range of terms, concepts, and principles from the different sciences and fields and in a way that is both clear and captivating.

Chapter 9 is of special interest to the analytic reader as it offers a case illustration of rehabilitation through psychoanalytic treatment. Here Doidge's point is that psychoanalysis, from its very start in Freud to the present day, is a treatment which is actually guided by an implicit recognition of the brain's plasticity and by attempts to facilitate the brain's potential for change.

Thus, while the topics of the book are varied, they all are aimed at demonstrating that successful treatments are actually founded on what we now know to be the brain's plasticity. More importantly, they are all used to support the claim that the newly found knowledge regarding this plasticity allows for cures that could have never been dreamt of in the past and explains existing cures. This knowledge has therefore initiated a kind of therapeutic revolution.

In our view, however, the book is also fundamentally flawed. This has to do with the very problematic nature of its theoretical assumptions and claims, implicit and explicit, regarding the relationship between the biological and the mental spheres, and invalid inferences made from findings in one sphere to the other. These problems have for the most part been overlooked in the many very positive reviews this book has received both in the general press¹ as well as in the supportive references to the book in analytic contexts (e.g. Fertuck, 2008; Frank, 2008)². In drawing attention to them here we will argue that Doidge is misguided in portraying the treatment innovations and improvements as being derived from or based on developments in the field of brain research.

The fact of the matter is that neuroscientific findings have had no meaningful impact on the nature of the treatments or their success.³ This becomes blatantly clear from a critical examination of the cases that Doidge presents to demonstrate this purported impact. One may presume that the dramatic changes that he describes involve changes in the brain, and neuroscientific techniques may be able to show this. The changes themselves, however, in no way rely on neuroscientific findings. Nor do retrospective

¹See the author's website: <http://www.normandoidge.com/normandoidge/REVIEWS.html>

²In Deutsch and Deutsch's review (2007) the authors acknowledge that the book has limitations, but nevertheless emphasize its overall importance for psychoanalytic thinking. Also, Doidge's book has been cited in psychoanalytic and psychotherapeutic papers where it is described as advancing psychoanalysis and psychotherapeutic thinking (see, for example, Hall, 2008, p. 847). A noted exception to this trend is a review by McManus (2009), a UCL Professor of Psychology and Medical Education who offers a very brief but scathing critique of the book. This critique and the kind of opposition it poses to neuropsychanalysis has not, however, been acknowledged within the analytic literature.

³As we shall see, unfounded beliefs regarding the value and relevance of neuroscientific findings may influence choice and course of treatment. But this points to the value and relevance of the beliefs rather than to the value and relevance of the findings themselves. To highlight that our study investigates the latter, at times, we refer to the '*meaningful*' impact of the findings and their '*fundamental*' relevance.

findings regarding the neuronal changes that took place provide information relevant to future treatment. Also, focusing on the way in which the book presents the tie between brain research and psychoanalytic theory and practice, we will show that the misapplication of neuroscientific ideas to the mental sphere shapes how clinical phenomena are regarded in a way that leaves little room for a truly psychoanalytic approach.

On wrongly attributing mental/functional improvement to theories or findings regarding neuroplasticity

Let us now examine some of Doidge's case studies to see the fundamental irrelevance of the neuroscientific findings to the therapeutic processes that he describes and how, despite this, he presents the findings as crucial to the therapy.

Some illustrations

In Chapter 2, entitled 'Building her own Brain', Doidge reports the fascinating case of Barbara Arrowsmith Young. As a child, Barbara was brilliant in some fields – her visual and auditory memory were in the 99th percentile – but there were some fields in which she was significantly limited. Doidge adds: "Because the part of her brain that helps to understand the relationships between symbols wasn't functioning normally, she had trouble understanding grammar, math concepts, logic and cause and effect" (p. 29). Barbara thought that these could not be improved.

At a certain point, on the advice of a friend, Barbara read about the research of Alexander Luria. Luria described someone who had similar mental difficulties to Barbara's although, in that person's case, the difficulties were derived from brain damage. Luria's research left Barbara feeling hopeless since it did not offer any course of treatment.

Barbara then heard of a study conducted by Mark Rosenzweig from the University of California at Berkeley. Doidge explains:

[He] had studied rats in stimulating and nonstimulating environments, and in post-mortem exams he found that the brains of the stimulated rats had more neurotransmitters, were heavier, and had better blood supply than those from the less stimulating environments.

(p. 35)

This study, according to Doidge's description of the course of events, led Barbara to attempt to change her brain structure and thus to change. While reading Rosenzweig's research "lightning struck" (*ibid.*). Linking Rosenzweig's research, which showed that activity could change the brain's structure, with Luria's research that showed that mental difficulties may be associated with brain damage, she initiated a rigorous regime of exercising her weaker mental capacities and the effort indeed led to dramatic improvement.

Doidge not only considers this a good use of neuroscientific findings; he also goes as far as to affirm that Barbara made an important discovery about the brain. In Doidge's words:

The scientists who make important discoveries about the brain are often those whose own brains are extraordinary, working on those whose brains are damaged. It is rare that the person who makes an important discovery is the one with the defect, but there are some exceptions. Barbara Arrowsmith Young is one of these.

(p. 27)

These conclusions are, however, completely unfounded. She neither made a discovery regarding the brain nor did she make real use of existing findings.

Barbara's reading about brain research apparently had an impact on her rehabilitation process, but it was not the neuroscientific contents of what she read that played the crucial role. It may be seen, for example, that the study that influenced her did not refer to someone with similar disabilities, and did not suggest any particular course for improving mental abilities. It did not even demonstrate that damaged brains could be rehabilitated or developed. It dealt with the normal brains (of rats) and showed that these brains develop in a stimulation-rich environment. It thus could not be an inference from the method or findings of the neuroscientific study *per se* that affected Barbara's course of action and its successful outcome.

It would seem, rather, that the study served as a general source of inspiration. It served for Barbara as a sort of motto: 'the brain could be modified'. With that in hand (not any new or old neuroscientific knowledge), she went ahead and devised her rigorous regime of exercises for her weakened functions which ultimately led to her improvement. To highlight the point: had Rosenzweig merely found that providing rats with stimulating environments improves their mental activities, without offering any information regarding the state of the rats' brains, Barbara would have had just as much reason to devise her regime. The fact regarding the brain does not play any directly meaningful role. Barbara may have been influenced emotionally by Rosenzweig's reference to changes in the brain, but the actual content of his findings regarding these changes are irrelevant to her situation and could not have had any effect.

It is important to note that the course of Barbara's treatment is not described as having been determined by information regarding a specific area of her brain. This is not surprising. As we have seen, the brain is irrelevant to the connections she made between the limitations of her mental functioning and the training she developed to overcome them. Even if Barbara were to have associated her limitations with a certain area of the brain (as Doidge does in a very general way), this would not have had clear-cut predictive implications for the effectiveness of the training programme. As is well known, (a) the same mental functioning could be determined by different areas of the brain and (b) training could lead to the development of capacities through affecting other areas of the brain without modifying the areas associated with the limitation (Levin and Grafman, 2000).

In fact, Doidge himself illustrates this point in another of the cases he describes, that of Michelle (Chapter 11). This case, too, provides evidence of the irrelevance of the neuroscientific findings to the therapeutic process. Michelle was born with only a right brain hemisphere. She suffered from

certain mental and physical developmental impairments but she overcame many of them and was able to develop mental functions usually associated with the left hemisphere. This occurred naturally, without any special training, and Michelle's family and doctors were unaware (for a number of years) of the fact that part of her brain was missing. A CAT scan performed when Michelle was 3½ years old made them aware of this fact. No expert on neuroplasticity was brought in to help at that point; however, her functioning continued to improve.

Although Michelle's case illustrates the brain's plasticity, it also illustrates how knowledge regarding the brain and its plasticity is irrelevant to the understanding of the process of mental development in cases like hers. It is clear from Doidge's description that in such cases, in which a substantial part of the brain is missing, one cannot predict which brain areas will take over which mental functions. This will depend to a large extent on the patient's life experiences and the ways in which the brain is activated by those experiences. Doidge, however, does not recognize the implications of this – if factors external to the brain significantly determine its functionality and the degree of its plasticity, this sets limits on what can be known from the study of the brain itself about how the brain will develop. Even more so, it limits what we can learn from such studies about mental functioning.

We turn now to yet another illustration of this. Doidge describes how Michael Merzenich, “the world's leading researcher on brain plasticity” (p. 46), and three associates “formed ... the nucleus of a company, ‘Scientific Learning’, that is wholly devoted to using neuroplastic research to help people rewire their brains” (p. 70). This company developed a “series of plasticity-based computer programs, *Fast ForWord*” for dealing with learning and language disabilities. According to Doidge, this software works on principles that are “consistent with the laws that govern brain plasticity” (p. 47).

When one examines Doidge's description of Merzenich's computer programs, however, it becomes apparent that, aside from the use of jargon associated with brain research, the programs have no tie to neuroscientific findings or principles of any kind. For example, we are told that *Fast ForWord* is disguised as a children's game that responds in a funny and encouraging way when the child being treated attains one of the goals that are set. Doidge explains the underlying neuroscientific principles thus:

This ‘reward’ is a crucial feature of the program, because each time the child is rewarded, his brain secretes such neurotransmitters as dopamine and acetylcholine, which help consolidate the map changes he has just made.

(p. 71)

This may indeed be an accurate description of neurological correlates of rewarding situations. Moreover, such jargon may impress parents considering the program, and the program may indeed be the most successful ever.⁴ But the idea that rewarding children is conducive to their learning is not a consequence of any neurological finding. It is known and validated indepen-

⁴While Doidge presents this program in a very positive light, other researchers have revealed a more mixed picture (Borman *et al.*, 2009; Gillam *et al.*, 2008).

dently of any knowledge of brain secretions or neural networks. In fact, any parent or teacher knows that positive responses may be motivating or arousing, and that, in general, a happy child is a child more willing and better able to learn. The principles involved are basic principles of learning, based on the study of the cognitive, psychological processes involved when a child learns – not neuroscientific ones.⁵

There is yet another flawed example that Doidge brings to demonstrate how neuroscientific findings significantly advance our understanding of mental processes. Doidge writes:

Merzenich and Jenkins also found that as neurons are trained and become more efficient, they can process faster. This means that the speed at which we think is itself plastic.

(p. 67)

But clearly the inference here is not from a neuroscientific finding to human activity. Rather, it is the other way around. We all know that when a person (not a neuron) practices a certain activity, including certain kinds of thinking, he becomes faster at it. This plasticity of thinking is an obvious behavioural fact and is not based on or derived from any neuroscientific findings provided by Merzenich and Jenkins. Based on this observed behavioural fact one may make inferences regarding the processing speed and efficiency of neurons.

In other words, here, too, the researchers do not discover plasticity through their neuroscientific studies as they and Doidge claim. They merely discover the neuronal correlates of an obvious and presupposed plasticity. As a result, the program that they propose is not derived in any significant, necessary way from neuroscientific evidence regarding the brain's plasticity.

This fallacy – whereby well-known or presupposed psychological and behavioural facts are presented as though they are findings derived from neuroscientific research regarding the brain's plasticity – permeates the book. We are led to believe that such research teaches us something new about vision, learning processes, or physiological capacities (such as the use of damaged limbs following stroke). However, for the same kinds of reasons described above, this is not at all the case. At the very most, what we learn from this research is about some neuroscientific correlates of mental and functional facts.

The negative impact of understanding in terms of neuroplasticity

The misguided use of brain research terminology that we have pointed to here may be appealing and inspirational to some. Unfortunately, it also has

⁵In analogy: we understand our preference for sugar in our coffee independently of the explanations, however interesting they may be, of the neurological effects of sugar. Moreover, if we would want to understand the experience of sweetness in a specific individual (as a private chef might) we would have no interest in general findings about brain correlates. We would rather be concerned with the individual's lived experience of different foods, e.g. how he reacts to the different foods we offer and what he says to his past experiences with food.

a negative impact on the way the mental sphere comes to be regarded (see Blass and Carmeli, 2007). This impact may be seen clearly throughout Doidge's book. In the following examples, we will first describe how the notion of plasticity is wrongly applied to the explanation of therapeutic processes. We will then show the broader negative effects of this misuse.

Chapter 6 deals with Obsessive Compulsive Disorder (OCD). After describing the disorder Doidge notes that medication and behavioural therapy are only partially helpful for many people. In contrast, we are told, a form of therapy developed by the UCLA psychiatrist Jeffrey M. Schwartz has been found to be effective. This therapy developed out of Schwartz's study of brain images of OCD patients and a notion of the brain's plasticity that emerged from it. Doidge states that this was "... the first time, to my knowledge, that such brain scans as the PET helped doctors both to understand a disorder and to develop a psychotherapy for it" (2007, p. 169).

The model of OCD and its treatment that Schwartz proposed is based on an examination of the psychological process that occurs when one makes a mistake, and of the correlates of that process in the brain. He points to three things that happen. First, we feel that we made a mistake. Brain imaging has shown that the detection of the mistake underlying this feeling occurs in the orbital frontal cortex. The more obsessive one is, the more activated one's orbital frontal cortex becomes. Second, we grow anxious, which leads us to try to correct the mistake. Brain imaging has shown that the cingulate gyrus, which is informed about the mistake by the orbital frontal cortex, triggers the dreadful anxiety that something bad is going to happen by sending signals "to both the gut and the heart (p. 169). Finally, when we have corrected the mistake, an automatic process occurs which allows us to move on to the next thought or action. The feeling of having made a mistake and the anxiety associated with it disappear. The area responsible for this shift is the caudate nucleus.

In people suffering from OCD this shift does not occur. Schwartz explains:

The orbital frontal cortex and the cingulate turn on and stay on as though locked in the 'on position' ... Because the caudate doesn't 'shift the gear' automatically, the orbital frontal cortex and the cingulate continue to fire off their signals, increasing the mistake feeling and the anxiety. Because the person has already corrected the mistake, these are, of course, false alarms. The malfunctioning caudate is probably overactive because it is stuck and is still being inundated with signals from the orbital frontal cortex.

(p. 170)

Schwartz's treatment is aimed at changing this problematic course of brain activity, this incapacity to shift gears which accounts for the OCD functioning, by releasing the over intense tie between the orbital frontal cortex and the cingulate and normalizing the activity of the caudate nucleus. The treatment is comprised of two steps. First the patient learns to re-label what is happening to him. He understands that his obsessive thoughts are not real (for example, he has not contracted an incurable disease), but rather that

they are an expression of an OCD episode. In this way he forms an emotional distance from them and can feel calm instead of anxious. As part of this he has to remind himself (in various ways) that the reason that the episode does not pass is the problematic connections in his brain. Focusing on the structure of the obsession rather than its content, according to Schwartz, is essential to this process.

The second step of this treatment involves the patient focusing on positive and pleasurable activities (for example, gardening, music, helping others) whenever an obsessive episode occurs. It is necessary to do something distracting and in this way initiate the gearshift. Repeated practice of initiating the shift creates flexibility in this mental capacity.

As in our previous illustrations, however, here too it may be seen that brain research did not actually determine this model or the treatment in any way. Any lay person who encountered someone with OCD would recognize his 'stuckness', his inability to let go of ideas even though they are mistaken. At best, what Schwartz adds here is information regarding these events' correlates in the brain. Moreover, the therapeutic method of suggesting to the patient that these brain correlates are responsible for the obsessive ideas is not in any way derived from brain research, but rather from psychological understanding of what allows for emotional distancing. The claim that in this case PET scans "helped doctors both to understand a disorder and to develop a psychotherapy for it" (Doidge, 2007, p. 169) is completely groundless.

Schwartz, as well as Doidge, doubts the efficacy of alternative therapies, pointing to their limited outcomes but more importantly to their limited logic. They oppose a behavioural approach, which involves exposure to anxiety arousing stimuli while preventing the obsessive-compulsive responses to them ('exposure and response prevention') because it tends to increase anxiety and for many patients arouses resistance. Thus it is helpful only to half of those who suffer from OCD and even for them only to a limited degree. Schwartz (and Doidge) also opposes cognitive approaches that encourage directly confronting the irrationality of the obsessive thoughts since the patients are aware of their irrationality. The problem is in the emotional sphere. To teach someone to say 'my hands are not dirty' will only involve him repeating something that he already knows, and will not affect his feeling that they are.

This critique of alternative treatments of OCD further highlights the irrelevance of neuroscience to Schwartz's treatment. For it may be seen that the therapeutic approaches that Schwartz rejects are just as based (or rather unbased) as his own on neuroscientific findings regarding plasticity. The providers of these treatments also think that the patient's problem is that he is stuck in his thoughts and actions and that their treatments offer him a way to let go of these. The therapies they offer also rest on the notion that the brain is plastic, that is, open to change. The therapies may differ from Schwartz's in their measured effectiveness, and they definitely do differ in regard to the method applied to bring about change. But these differences arise from different understandings of the kinds of human interactions that

create emotional distance and relieve 'stuckness'. These are understandings of a psychological rather than a neuroscientific kind.

In the context of alternative treatments, Doidge also mentions the psychoanalytic treatment of OCD. He writes:

Psychoanalysts too have focused on the content of the symptoms, many of which deal with troubling sexual and aggressive ideas. They have found that an obsessive thought, such as "I will hurt my child", might express a suppressed anger at the child, and that this insight might, in mild cases, be enough to make an obsession go away. But this often does not work with moderate or severe OCD. And while Schwartz believes that the origins of many obsessions relate to the kind of conflicts about sex, aggression, and guilt that Freud emphasized, these conflicts explain only the content, not the form of the disorder.

(p. 172)

It is here that we may begin to see that the brain research adduced in favour of Schwartz's treatment is not only, as we have shown, irrelevant to it, but also has a negative impact on our capacity to understand the ways in which disorders may be treated. This is because Schwartz's focus on neural correlates precludes the possibility of ever coming up with a content-oriented explanation of OCD. Areas in the brain are associated with functions, not with contents. Recognizing the involvement of a certain area of the brain may tell us that we are doing a mathematical calculation, but it cannot tell us what the numbers involved in it are. It may tell us that we are thinking of something frightening, but not that the frightening thought is of white wolves rather than a big black spider. This is not a limitation of the state of the art of neuroscience, but inherent to the possibility of inferring from biological states to specific thoughts (see Blass and Carmeli, 2007, p. 25).

Thus, while therapists and researchers such as Doidge may regard Schwartz's brain research as evidence for a functional explanation of OCD, for the preference of a form-oriented model over one of content, the fact of the matter is that no such evidence is provided. Schwartz does not *find* that form is more important than content, but rather, in choosing a method of investigation that examines neural correlates of OCD, he is limiting himself to formal explanations; he is assuming rather than finding that they are preferable. The value of content-oriented explanations are automatically downplayed; they are regarded as possibly interesting, but not really a sound basis for an effective cure, which from the neuroscientific standpoint would have to be a form-oriented one.

The fact that content-oriented explanations are given short shrift is particularly strange in the context of OCD. This is because, while the OCD patient may be more inclined to get stuck than others, he does not get equally stuck on all ideas. Clearly, obsessive-compulsive tendencies tend to find expression especially in relation to certain contents. So, *prima facie*, there seems to be very good reason to pay attention to content. To fail to do so would be like explaining one's love for a specific painting in terms of a highly developed area in the brain that corresponds to an aesthetic capacity. Such an explanation would be blatantly flawed since it would not

explain why *this* painting is the object of the desire. To explain the specific desire it would make sense to take into account something having to do with the content of the painting.

It may also be seen that, in following the neuroscientific focus on form rather than content, Doidge adopts a certain notion of what constitutes a positive outcome of treatment – only symptom reduction, behavioural change. Other kinds of treatment outcomes, such as self-integration or self-understanding, require a concern with meaning. Here the opposition between Doidge and the psychoanalytic perspective is pronounced. Psychoanalysis is not only concerned with meanings and their integration, but rather from a psychoanalytic perspective it would be wrong to steer the patient away from content, or even further, as Schwartz does in the case of obsessive thoughts and activities, to try to convince the patient that what goes on in his mind and what he does are meaningless. It is part of the analytic task to understand these self-expressions and to do so in a way that allows the patient to think, rather than be taught how he should be thinking.

The differences between Doidge's and Schwartz's view and that of psychoanalysis are not due to any research findings. Rather, they reflect a choice about what constitutes valuable therapeutic development. The fact that this is a choice gets lost when the focus on form rather than content is presented as the approach better grounded in the hard science of brain research. This is yet another detrimental effect of brain research on the understanding and treatment of mental states.

Psychoanalysis and brain plasticity

The chapter that Doidge devotes specifically to psychoanalysis offers the most striking example of negative consequences entailed by thinking of human experience in neuroscientific terms. Because of its specific focus on psychoanalysis, we will discuss this chapter in greater detail. First, we will show how Doidge's claims regarding the relevance of neuroplasticity to psychoanalysis are mistaken. We then discuss the negative consequences of maintaining these mistaken claims.

In attempting to establish the connection between psychoanalysis and brain research, Doidge relies heavily on a series of studies conducted by the Nobel Prize laureate Eric Kandel. In one study, Kandel revealed the neural correlates of learning processes (for example, learning to avoid shock) in giant marine snails of the genus *Aplysia*. He went on to show how the snails develop long- and short-term memory. Later, together with molecular biologist James Schwartz, he revealed the molecular processes involved in the development of long-term memory in these snails. Doidge writes:

They showed that in the snails, for short-term memories to become long-term, a new protein had to be made in the cell. The team showed that a short-term memory becomes long-term when a chemical in the neuron, called protein kinase A, moves from the body of the neuron into its nucleus, where genes are stored. The protein turns on a gene to make a protein that alters the structure of the nerve ending, so that it grows new connections between the neurons. Then Kandel, Carew,

and colleagues Mary Chen and Craig Bailey showed that when a single neuron develops a long term memory for sensitization, it might go from having 1300 to 2700 synaptic connections, a staggering amount of neuroplastic change.

(2007, p. 220)

Doidge then presents the conclusion drawn from this study. This conclusion regards the effects of learning on the expression of our genes:

The same process occurs in humans. When we learn, we alter which genes in our neurons are 'expressed,' or turned on.... Most people assume that our genes shape us – our behavior and our brain anatomy. Kandel's work shows that when we learn our minds also affect which genes in our neurons are transcribed. Thus we can shape our genes, which in turn shape our brain's microscopic anatomy.

(p. 220)

Doidge then ties this to the process of psychotherapy. He writes:

Kandel argues that when psychotherapy changes people, "it presumably does so through learning, by producing changes in gene expression that alter the strength of synaptic connections, and structural changes that alter the anatomical pattern of interconnections between nerve cells of the brain." Psychotherapy works by going deep into the brain and its neurons and changing their structure by turning on the right genes.

(p. 221)

It is here that the assumed link between psychoanalysis and neuroscience emerges. The analyst is viewed as one who recognizes the plasticity of the brain and, through the analytic treatment, attempts to alter neuronal networks. He is, so to speak, a "microsurgeon of the mind" (p. 221).

Following up on Kandel's insight, Doidge tries to show that Freud himself emphasized the plastic nature of the brain and had knowledge in this context that only much later was formally discovered by neuroscience. For example, he writes that: "The first plastic concept Freud developed is the law that neurons that fire together wire together" (p. 221). Similarly, Doidge says that: "To be changed, Freud argued, memories had to be conscious and become the focus of our attention, as neuroscientists have since shown" (p. 225). "Most important", Doidge explains:

Freud discovered that transference of early traumatic scenes could often be altered if he pointed out to the patient what was happening when the transference was activated and the patient was paying close attention. Thus, the underlying neuronal networks, and the associated memories, could be re-transcribed and changed.

(*ibid.*)

Doidge goes on to present not only Freud, but psychoanalytic theory in general, as supportive of the notion of the plasticity of the brain.

While the detailed quotes from Kandel's work in this context imply a close tie between psychoanalysis and neuroscience, the fact is that the connections are ambiguous and superficial. What Kandel's studies clarified are the neurological processes that are involved in memory formation and

change and in the processes of learning. One could argue that psychotherapy entails a learning process and changes in regard to memories and therefore also changes in the brain. But this tells us nothing meaningful about psychoanalysis, nor does it point to how a neuroscientific approach affected Freud's psychoanalytic thinking, as Doidge claims it did.

Doidge stresses correlations between Freud's analytic ideas and those of neuroscientists. For example, both recognize the importance of making memories conscious in order to facilitate change. However, these correlations do not point to significant connections between the fields. Neuroscientists may recognize that memories have to be made conscious in order to be changed (whatever that may mean), but this is not a neuroscientific finding. So the fact that Freud also recognized this has no bearing on his relationship to neuroscience.

To use an analogy: neuroscientists may recognize that one needs water to live, but this does not mean that all those who recognize this necessity are neuroscientists or support neuroscience. Similarly, neuroscientists may choose to describe the neuronal correlates of psychological phenomena that Freud identified, such as transference, but this could not possibly be regarded as evidence of Freud's affiliation with neuroscience. If it were, and one accepts that all psychological phenomena have neuronal correlates, then all psychologists would be, by definition, equally affiliated with neuroscience. In other words, it makes no sense to point to a special affiliation of a psychological approach with neuroscience on the basis of the fact that the phenomena it describes have neuronal correlates. One may assume that all psychological phenomena have such correlates.

Another way in which Doidge ties psychoanalysis to neuroscience is through pointing to the neuronal effects of psychoanalysis.⁶ In this context he cites Kandel's statement that: "There is no longer any doubt ... that psychotherapy can result in detectable changes in the brain" (p. 233). But here too one must question whether such effects point to a significant tie. For if indeed all psychological events have neuronal correlates, then any psychological change would involve a change on the neuronal level. The question would only be whether or not it is identified. Moreover, *pace* Doidge, neuroscience's ability to identify such changes following psychoanalysis contributes little to its validation as a form of treatment. If a psychological change took place following analysis and no neuronal correlates were found, we would not deny the effect; rather, we would assume that the available technology failed to identify it. Conversely, if changes in the brain were noticed after treatment despite the fact that no meaningful psychological change could be seen, we would hardly call it a success. Ultimately, what matters in determining the effectiveness of a treatment is the psychological, not the neurological, level of change.

Doidge, through a case report, further develops and illustrates his thinking regarding the relevance of neuroscience (and specifically its findings

⁶It should be noted that the evidence that Doidge brings in this context are from studies of interpersonal psychotherapy based on the approaches of Bowlby and Sullivan. And one may question whether it is legitimate on the basis of such studies to draw conclusions regarding the effects of psychoanalysis.

regarding plasticity) to psychoanalysis. It is helpful to take a closer look at this case since it allows us to gain insight into the negative impact of turning to neuroscience in the effort to understand mental phenomena. The case is that of Mr L, a man in his 50s, incapable of forming significant emotional ties. As soon as he would get romantically involved, he would start backing off, feeling that: "There is a better woman elsewhere whom I'm being denied" (p. 216). He had been unfaithful to his wife and as a result lost his marriage, an outcome he greatly regretted. At the same time, he respected his wife and was not sure why he was unfaithful to her. And, although he was devoted to his children, he felt attached to them through a sense of duty rather than joyful affection.

Doidge describes Mr L's difficult background. His mother died when he was just over 2 years old, while giving birth to his sister. He then became ill and was sent off to be looked after by relatives. Although he had suffered a series of traumatic losses, Mr L did not think that these events had any real influence on his life and did not feel any sadness about them. He had no memories from age 4 or earlier, remembered little from his teenage years, and spoke as though nothing that had happened to him had registered. Nevertheless, there were clues that his losses did register: as he told his story it seemed as though he was still in shock, and he was haunted by dreams in which he was always searching for something.

To explain Mr L's predicament, Doidge turns to developmental research. He suggests that:

[An] important critical period lasts from approximately ten or twelve months to sixteen or eighteen months, during which a key area of the right frontal lobe is developing and shaping the brain circuits that will allow infants both to maintain human attachments and to regulate their emotions. This maturing area ... is called the *right orbitofrontal system*.

(p. 226)

Doidge describes this developmental process on two levels, the first interpersonal and the second neuronal. This allows him to offer the following explanation:

For children to know and regulate their emotions, and be socially connected, they need to experience this kind of interaction many hundreds of times in the critical period and then to have it reinforced later in life. Mr L lost his mother only a few months after he completed development of his orbitofrontal system. So it fell on others, who were themselves grieving and probably less attuned to him than his mother would have been, to help him use and exercise his orbitofrontal system, lest it begin to weaken... If others cannot help him soothe himself and regulate his own emotions as his mother did, he learns to 'autoregulate' by turning off his emotions. When Mr L sought treatment, he still had this tendency to turn emotions off and trouble maintaining attachments.

(p. 227)

It should be clear by now that this kind of emphasis of neuronal correlates of phenomena that are known through psychological study adds nothing to our understanding. For example, we know because of infant observation,

not the observation of brains, when what is considered to be normal development takes place.

But beyond the irrelevance of the turn to neuroscience, this emphasis also constricts Doidge's ability to think about the patient, as may be seen in the clinical material he goes on to present. Neuroscientific models constrict analytic thinking in two major ways. First, and as we already saw above, it leads the analyst to focus on mental functions to which he can find brain correlates. In this way he will tend to think about his patient's problem in terms of deficits – for example, the absence of some emotional ability as a consequence of an undeveloped brain area – and ignore complicated conflicts between meaningful ideas (see Deutsch and Deutsch, 2007). Second, even when focused on content rather than the form of his patient's words, he will tend to understand his problems in terms of simple associations between well-defined ideas. This is because the subject of neuroscience being physical matter is concerned as it is with well-defined and discrete observable events. It cannot address complex associations between ideas, symbols and phantasies, because these rely on contexts of meaning that are holistic rather than discrete in nature (see Davidson, 1980). Consequently, an analyst inspired by neuroscientific research will seek similarly simple explanations and will be predisposed to make connections only between discrete phenomena, such as between two cognitive functions. His interpretations too will be focused on well-defined and discrete phenomena that will render them almost behaviouristic in nature.

Let us see how this plays out in Doidge's analytic case. Doidge describes Mr L's emotional life and his actions as consequences of (a) developmental deficits that obstructed the formation of important emotional functions and (b) associations that developed between certain life events and the experience of a state of threat and anxiety. With this view that emphasizes the direct impact of real life events, the transference that evolves is also viewed as a realistically positive one, with the analyst actually providing the missing maternal functions and in this way furthering the developmental process. Doidge writes:

Past and present were being mixed up together, and a transference was being activated. At this point I, as an analyst, did what an attuned mother does, when she develops the orbitofrontal system, by pointing out emotional 'basics' – helping him name his emotions, their triggers, and how they influenced his mental and bodily states. Soon, he was able to spot the triggers and emotions himself.

(2007, p. 230)

Doidge explains that in the transference the brain recognizes the early maternal pattern of relationship, a view that fits nicely with simple neuroscientific models, but leaves no room for the idea that internal phantasies are finding expression in the transference. Indeed, no such ideas are discussed by Doidge – not even in a theoretical context.

The simple, observable approach of neuroscience also finds expression in Doidge's view of memory. Guided by neuroscientific findings according to which trauma damages the hippocampus and the ability to form explicit

long-term memories, Doidge suggests that Mr L's limited explicit memory from his early years and adolescence is due to a shrinking of the hippocampus following trauma. When Mr. L was in his 20s, Doidge claims, the hippocampus regrew, apparently as a result of improvement in relation to Mr L's depression. This, according to Doidge, explains why Mr L has memories from that age onwards.

The idea that repression may be responsible for Mr L's limited memory is not raised as a possibility. Doidge immediately adopts the physical damage option. It is not presented as a hypothesis, or one possibility among other more dynamic explanations, but rather is offered as real and certain. But, of course, whether events were never registered or are being repressed could hardly be a matter of certainty (see Blass and Carmeli, 2007, pp. 24–5).

To illustrate the problem: Doidge acknowledges that one of Mr L's few memories is seeing his mother at her funeral, but he knows this only from some associations to a dream. It seems that, if Mr L had not mentioned this, Doidge would have concluded (wrongly in his own view) that no memories of the funeral were registered due to damage to the hippocampus. Moreover, Doidge's focus on physical damage also leaves no room for the question of the *meaning* of Mr L's memories. Only the fact of their presence or absence is of concern to Doidge. Here, too, Doidge abandons the complexity of analytic thinking and practice.

The choice of simple and physically oriented explanations is particularly notable in Doidge's description of the role of the dream in psychoanalytic thinking. He writes:

Why are dreams so important in analysis, and what is their relationship to plastic change? Patients are often haunted by recurring dreams of their traumas and awaken in terror. As long as they remain ill, these dreams don't change their basic structure. The neural network that represents the trauma ... is persistently reactivated ... Should these traumatized patients get better, these nightmares gradually become less frightening ...

(p. 238)

In other words, dreams are central to psychoanalysis because of their role in representing trauma. The neglect of the wealth of psychoanalytic thinking regarding the dream is striking.

Such neglect becomes even more striking in Doidge's interpretations of his patient's dreams. The dreams' contents are interpreted as direct and self-evident symbols of early traumatic events, which are only mildly disguised. For example, Mr L reports regarding one of his dreams: "I am searching for something, I know not what, an unidentified object, maybe a toy, which is beyond familiar territory ... I'd like it back again" (p. 217). Reflecting on this, Doidge states that, although Mr L comments on the fact "that the dream represented 'a terrible loss' ... remarkably, he did not link it to the loss of his mother or family" (p. 217). For Doidge the link is obvious. Mr L's failure to take note of it is, therefore, remarkable. More complex associative possibilities are not explored.

Another blatant example of Doidge's constricted perspective on the clinical material may be seen in relation to the final dream brought in the analysis, the one regarding the funeral of Mr L's mother. Mr L says: "I saw my mother playing the piano, and then I go to get someone, and when I return, she is in a coffin" (p. 237). This dream led Mr L to an association that he felt to be an actual memory of the situation in which he saw his mother in her coffin. Consequently, he felt the loss with great intensity: "He let out a loud wail and, overcome with primitive grieving, his whole body convulsed for ten minutes" (p. 237).

At this point, Doidge wonders if the events described were actually remembered or whether they were fantasized, based on the patient's wish that he could remember the event that he experienced at the age of 26 months. Doidge does not reject outright the possibility of fantasy being involved here, but he is quite certain of it being an actual memory. In support, Doidge mentions that a relative confirmed that the patient was present at the funeral and that the coffin was open and presents research that memory at that age is possible. Once again, for Doidge the central issue is the question of the reality of the memory.

In sum, we have seen here how the attempt to integrate neuroscience with psychoanalysis has a negative impact on the appreciation and utilization of psychoanalytic thinking and practice. Doidge's understanding of analytic patients can centre only on deficits, not conflict, since this is what the neuroscientific models can allow for and encourage (see Deutsch and Deutsch, 2007, p. 351). Freud's complex thinking on the dream, which encompassed ideas on repressed wishes and conflicts and their relationship with various aspects of the individual's life and forms of processes, is replaced by the extremely simple and one-dimensional view of the dream as a method for dealing with trauma. Again, this shift is not based on an in-depth critique of Freudian theory or practice (which may be legitimate) but simply on the basis of the adoption of a model of explanation drawn from the world of neuroscience. It is when such models which, as we have argued, cannot encompass content or complexity are applied without warrant to the realm of the understanding of the meanings and motives of human thought, feeling and action, that analytic thinking and practice becomes deprived of what is most essential to it.

Although Doidge surprisingly does not mention it, this chapter's title, 'Turning Our Ghosts into Ancestors', is an oft-cited metaphor introduced by Hans Loewald in his 1960 paper, 'On the therapeutic action of psychoanalysis'. There, Loewald described the process of change that takes place through working with the transference. He wrote:

In the daylight of analysis the ghosts of the unconscious are laid and led to rest as ancestors whose power is taken over and transformed into the newer intensity of present life, of the secondary process and contemporary objects.

(Loewald, 1960, p. 29)

The transformations Loewald described involve complex psychic processes. Sadly, in an effort to portray analysis as a neuroplastic treatment supported

by brain research, Doidge's chapter on psychoanalysis severely constricts and distorts not only the conceptualization of these processes, but also the nature of psychoanalysis itself.

Concluding reflections

In this paper, we examined recent claims that evidence regarding the brain's neuroplasticity can significantly influence psychological understanding and development and is very relevant for psychoanalysis. Through a close study of Norman Doidge's book *The Brain that Changes Itself*, we showed that these neuroscientific findings are irrelevant to the psychological and psychoanalytical processes described. Moreover, we pointed out how regarding such findings as relevant to psychological and psychoanalytic processes poses an obstacle to the development of in-depth understanding. As we have seen, when one thinks in terms of the neuroscientific correlates of psychological processes, dimensions of meaning and content are ignored as factors determining psychological states. This is because the neuroscientific data can at best be shown to correlate with broad formal dimensions, such as the presence of certain kinds of affective states, or the activation of a certain kind of mental mechanism, but not with specific ideas. It may indicate the presence of thought processes associated with mathematics, but not the numbers of which a person is thinking. And it would never be able to tell us of the infinite range of associations a person has with those ideas, and the complex ways in which they interrelate in our minds and actions.

If one holds that psychoanalysis is defined by its concern with meaning and that its view of pathology and development is intimately tied to self-understanding and integration, one must conclude that it stands diametrically opposed to treatments which adopt the formal focus of neuroscience. For Doidge, there is no such opposition; but accordingly his view of psychoanalysis is a very limited one. It deals with deficits and with general and simple events, but not with the complex, multidimensional and ever-changing world of meaning, phantasy and conflict.

Another negative consequence of wrongly considering neuroscience to be relevant to psychological change is that it supports the illusion that knowledge of the brain and its plasticity could be therapeutic. Thus we are invited to devote our resources and energy to biological inquiry, while in fact this cannot help.

Pointing to the irrelevance of the findings on neuroplasticity and their harm further grounds ideas we put forth in our earlier critique of neuropsychanalysis (Blass and Carmeli, 2007). In that earlier work we addressed studies which argued that neuroscientific knowledge could be valuable to psychoanalysis because it provides information regarding dimensions of the person that are concrete and stable in nature, in contrast to the amorphous and changing nature of psychological phenomena. Here we addressed studies that value the contribution of neuroscience to psychoanalysis because it provides information regarding the brain's unstable nature, its plasticity. While the plasticity perspective is more attuned to ongoing environmental

influences on the brain, it fails to recognize that these influences can take infinite possible forms, the effects of which cannot be foreseen. It also fails to recognize that, as a consequence of this complexity, what can be learned from the study of the brain *per se* is sharply limited. Thus, what neuroscience primarily adds to psychological understanding is only anecdotal information regarding corollaries to psychologically described processes.

In the case of neuroplasticity, this anecdotal level seems especially trivial. It is a kind of translation to biological language of what is known by commonsense. We all know that people learn. Referring to this process as one of training the brain adds nothing. People change. To say that they are expressing neuroplasticity adds nothing to our psychological understanding. Similarly, it is meaningless to say that because Freud knew that people change, that understanding and experience modify the way one feels and acts, he offered 'neuroplastic therapy'. Chris McManus, Professor of Psychology and Medical Education at University College London, emphasizes this point in the sole strikingly critical review of Doidge's book (the review appeared in *The Psychologist*). McManus quips: "Should your behaviour change as a result of reading this review, then perhaps I also [like Freud] am a 'neuroplastician'" (2009, p. 40).

Given the triviality of translating psychological phenomena into the language of plasticity, some explanation of its popularity is in order. In our earlier work we tied its appeal to the biologicistic ethos of our contemporary culture, the view that what is real is biological (Blass and Carmeli, 2007, p. 36; see also Brothers, 2002, p. 869). The enthusiastic reception of Doidge's book in the popular press highlights this cultural trend, as may be seen in an excerpt from *The New York Times* review, which appears on the book's cover. It reads: "The power of positive thinking finally gains scientific credibility".

McManus maintains that the kind of brain-based analysis of psychological functioning and processes presented in Doidge's book – "neurobabble" as he refers to it – should only be of interest to psychologists if they are "sexing up grant applications, or persuading sceptical university administrators that psychology is actually serious science" (2009, p. 40; Brothers makes a similar point [2002, p. 869]). It seems, however, that in the course of the search for recognition and support, many psychologists and psychoanalysts have actually come to believe that being able to refer to neuroscientific correlates of psychological phenomena is indeed what makes their profession scientifically respectable. Perhaps others know better, but tolerate the neuropsychanalytic discourse so that psychoanalysis may benefit from the appreciation that neuropsychanalysis's purported scientific status supposedly endows. The consequence has been that, up until now, there has been no serious criticism of Doidge's book in the psychoanalytic literature.

As we have made clear in the course of this paper, such tolerance and support of the approach that Doidge's book presents are problematic, particularly in light of its negative implications for how psychoanalysis is understood and practised.

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